

Auditing a Transcriptomic Prognostic Funnel in Clear Cell Renal Cell Carcinoma

HYDRA-ccRCC development manuscript

Internal development draft. Not for direct submission.

Abstract

The one-gene rule failed its prespecified test. On the 517-patient nested cohort, adding the training-selected gene changed concordance by 0.0039 (patient-bootstrap 95% interval -0.0088 to 0.0171, conditional on the saved out-of-fold predictions). The point estimate is 0.0039. The interval crosses zero and reaches 0.0171, so the estimate sits below the prespecified +0.01 bar while the interval does not. That failure is not evidence that the 23 associations are null.

A ridge model of the reproducible genes, on those same patients and splits, changed concordance by 0.0257 (interval 0.0116 to 0.0413). Preprocessing and λ were taken from the outer training fold; the median selected λ was 40.22. The three-year Brier interval still crossed zero. The ridge arm is not a validated signature, and it was not fit on GSE29609 or E-MTAB-1980. The largest marked bar in the opening figure is this arm. It was not the acceptance test.

The external cohorts disagree, and both were inspected before this reanalysis. GSE29609 retained the TCGA direction for 5 of 21 mapped genes and showed FDR reversals for DDC and TCIRG1. E-MTAB-1980 retained it for 21 of 22, with strict support for 12. That agreement is not external validation. Alternate aliquot rules move the edge of the 23-gene list. The unpaired GSE53757 fit kept every high-confidence gene and does not prove that the inferred pairs are real patients. TRACERx agreement across genes was 98.5% [76.6–100.0] with the patient subset fixed and 86.3% [77.8–93.1] when membership was redrawn. None of these results is a clinical action.

HYDRA-ccRCC: Evidence-Hardening Funnel and Held-Out Validation

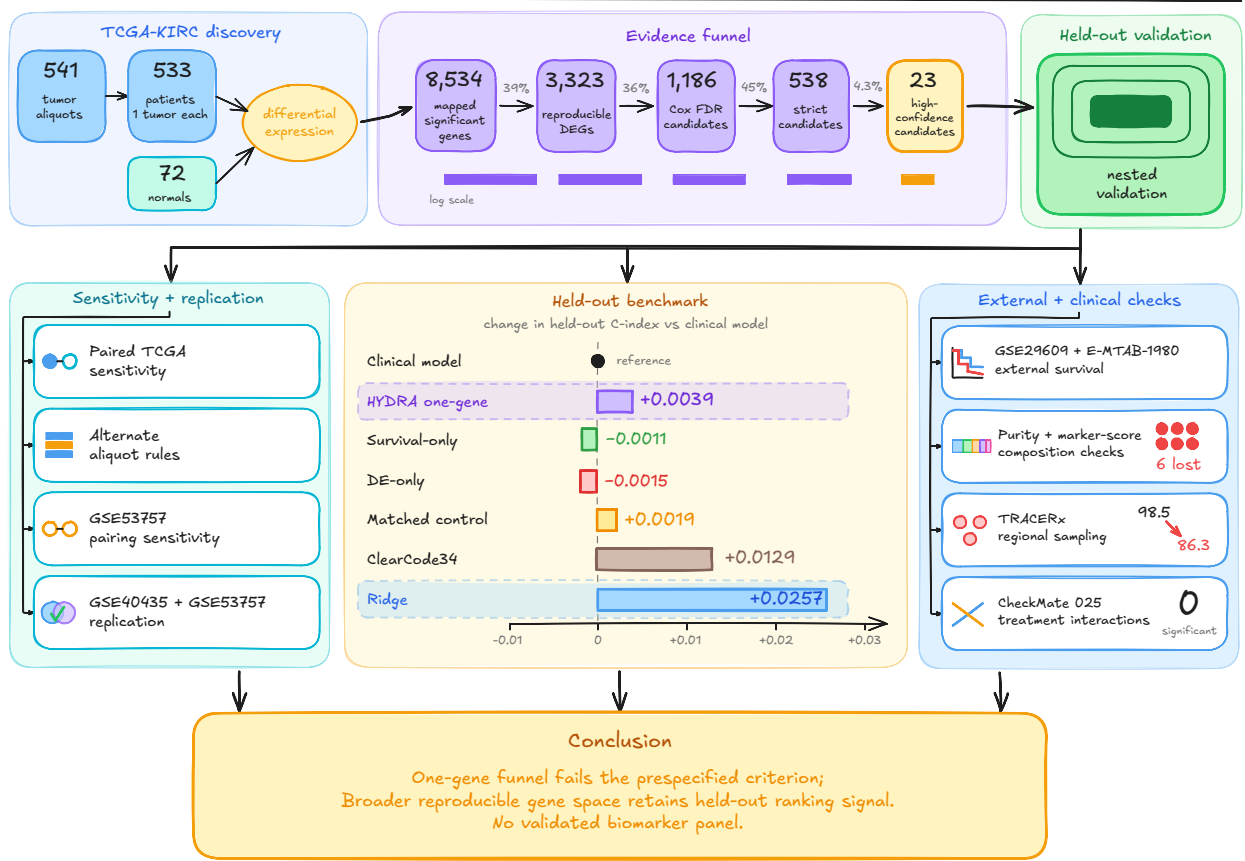


Figure 1: Evidence-hardening overview. The funnel runs from 541 tumor aliquots to 23 high-confidence candidates. Held-out changes are versus a clinical model. The dashed threshold is the prespecified $+0.01$ gain for the one-gene acceptance test. That test is the HYDRA arm. Ridge shows held-out ranking information in the reproducible set on these TCGA splits. It does not pass the one-gene test, and it is not a validated signature. External, composition, TRACERx, and CheckMate 025 checks are exploratory.

1 Introduction

Clear cell renal cell carcinoma (ccRCC) has broad metabolic, vascular, and immune transcriptional changes [4, 12]. A tumor-normal expression difference does not establish independent prognostic value: a signal can arise from the composition of the tissue, and repeated samples from one patient can make survival evidence appear more precise. We audited the public-data HYDRA-ccRCC funnel for these failure modes. The question is whether the one-gene rule generalizes, and whether the broader reproducible set still ranks patients when that rule does not.

2 Methods

2.1 Inputs, patient identity, and discovery

TCGA-KIRC STAR unstranded counts and clinical outcomes were retrieved through TCGAbiolinks [5]. For each patient and sample type, we selected the aliquot with greatest raw-count library depth and broke barcode ties so the choice is deterministic rather than biological. The 541 tumor aliquots came from 533 patients, including 175 deaths: four patients contributed three aliquots each, which accounts for all eight extra aliquots. Of those 533 patients, 16 are absent from the primary Cox model and the nested splits

because each fails exactly one required field—8 lack grade, 3 lack stage, 1 lacks age, and 4 have overall-survival time recorded as zero—leaving 517 patients and 170 deaths (5 of the excluded patients died). The stage-only sensitivity restores the grade-missing patients (525) and the grade-only sensitivity restores the stage-missing patients (520). All 72 normals were already unique by patient. Genes are matched by symbol after a dataset-specific collapse, recorded in the gene-mapping table. TCGA strips the Ensembl version, maps the id with org.Hs.eg.db, and keeps the first symbol. GSE40435, GSE53757, and GSE29609 take the first token of the platform gene-symbol field and average probes that share it. E-MTAB-1980 maps RefSeq identifiers the same way and averages rows that share a symbol. TRACERx matches candidate symbols to expression rownames. CheckMate 025 averages rows that share `gene_name`. Survival-event coding was checked against vital status and the appropriate death or follow-up date; joins were required to remain one row per patient. Three alternate gates, reported as sensitivities, dropped those four patients, kept the first barcode, or summed a patient’s tumor counts. Original retrieval dates for cached public inputs were unavailable, so the repository records source URLs and hashes without assigning new access dates.

DESeq2 compared the selected tumors and normals after requiring at least ten counts in ten samples [9]. A separate patient-blocked DESeq2 fit, with design patient plus tissue type, used the 72 selected tumor-normal pairs. GSE40435 and GSE53757 were analyzed with patient-blocked limma; each parsed patient was required to have exactly one tumor and one normal, and surrogate-variable analysis protected the paired contrast [8, 10]. GSE53757 has no field that stores one shared patient id for every pair. Adjacent rows were opposite tissues and the same stage in all 72 pairs, and 4 pairs also shared a title token, so the primary pair ids follow that adjacent order. An unpaired limma fit is a sensitivity. A reproducible DEG required TCGA FDR below 0.05 and absolute $\log_2$ fold change at least one, the same sign in both GEO cohorts, and nominal support in at least one, defined as a two-sided unadjusted $p < 0.05$ with that same sign. A separate apegglm analysis tested all count-QC genes without the fold-change gate; it was not allowed to redefine the primary candidate set [13].

The primary Cox model used continuous standardized expression, age, sex, categorical AJCC stage with Stage I as the reference, and grade collapsed to G1+G2 versus G3+G4 among complete cases [6]. G1 contains 14 patients in the TCGA clinical table, so it was not kept as its own level. The strict gate also required main-model FDR below 0.05, absolute log hazard ratio at least $\log(1.25)$, both GEO effects at least 0.25 in absolute $\log_2$ units, and concordant nominal support in two sensitivity fits: expression plus age, sex, and stage, and expression plus age, sex, and grade. The high-confidence gate required FDR below 0.01 and absolute log hazard ratio at least $\log(1.5)$. Those magnitude cutoffs are prespecified project thresholds, not a cited minimal clinically important difference. False-discovery correction stays inside one family at a time: differential expression, each GEO cohort, Cox models among reproducible genes, and the high-confidence tests for shape, composition, and treatment interaction. Proportional-hazards tests were diagnostics, never selection gates. The selected gene is the one with the highest evidence score, $-\log_{10}(\text{FDR}) + |\log \text{HR}|$ plus bounded fold-change terms, with remaining ties broken by FDR and then gene id. The acceptance test adds that one gene. The ridge fit is a separate multigene description. It does not replace the failed one-gene criterion or freeze a signature. It is a glmnet Cox model with Efron ties, clinical covariates unpenalized, and an L2 penalty ($\alpha = 0$) on every training-fold reproducible gene with nonzero training standard deviation. The penalty does not zero coefficients and does not name a gene subset. Before any penalty is chosen, the outer training fold fixes the design matrix: DESeq2 size factors are estimated on the training samples, test size factors use that training geometric-mean reference, expression is $\log_2(\text{count}/\text{size factor} + 1)$, and genes with zero training standard deviation are dropped. Clinical factor levels come from the 517-patient complete-case cohort; the values in a fit come from its training rows. λ is then chosen by an inner cross-validation that uses only those training patients. `cv.glmnet` first fits the λ path on all outer-training rows, scores mean partial-likelihood deviance on five event-stratified inner folds of those same rows, and keeps $\lambda_{\min}$. The criterion is deviance, not the concordance index and not the one-standard-error rule. Each inner fit standardizes its own training columns. The coefficients applied to the outer test rows come from the refit on the full outer training fold, and that fold’s column means and standard deviations are applied to the test rows. Outer test outcomes are not an input. The inner-fold seed is the project seed plus 31000 + 10 × repeat + fold, drawn before `cv.glmnet`. The reported contrast compares this joint fit with a separate clinical Cox model. Clinical coefficients inside the ridge fit are estimated together with the penalized genes, so the concordance increment is not an isolated gene effect and does not identify which genes produce the gain.

2.2 Selection-aware prediction and null simulation

Ten repeats of five event-stratified patient folds reran TCGA DE and every candidate gate within each training fold of the 517 complete-case patients defined above. Fold labels were redrawn for each repeat. Event stratification shuffled deaths and censored patients separately, and every outer test fold contained 34 deaths. The full-data stage-only and grade-only sensitivities contain 525 and 520 patients, respectively, and are therefore not numerically identical to these nested fits. GEO expression evidence remained fixed; test patients and their normals were excluded from training. That is the scope of “selection-aware” here: TCGA differential expression, the fixed GEO gate, the Cox gates, and the choice of one gene. It does not re-split the GEO cohorts or re-choose the 0.01 rule inside a fold. Library-size normalization and expression scaling were estimated on training samples and applied to held-out tumors. The nested fit used $\log_2$ normalized counts, while the full-data survival analysis used a variance-stabilizing transformation; this preserves training-only preprocessing but changes the expression scale, limiting direct coefficient comparisons. DESeq2 retained the primary outlier-replacement setting, with a logged no-replacement retry only if the pinned runtime failed in that step. The top qualifying gene was added to a clinical Cox model; when no gene qualified, its prediction equaled the clinical-only prediction. Out-of-fold concordance, three-year inverse-probability-of-censoring-weighted Brier score, and risk-score calibration slope were computed per repeat. The concordance increment received a patient-resampled 95% interval conditional on the saved out-of-fold predictions; it does not include variation from refitting selection or redrawing folds. The prespecified success rule required an increment at least 0.01, a conditional interval above zero, and a mean Brier difference no greater than zero. The 0.01 increment is this project’s size requirement, not a universal clinical standard. A lower Brier score is not, by itself, better calibration: discrimination and calibration contribute separately to the Brier score.

Two hundred clinical-only null datasets used an exponential baseline calibrated to the clinical Cox cumulative hazard with resampled censoring; each reran the full training selection on one held-out fold, cycling across the five fixed folds from repeat 1. This is a parametric clinical-risk null, not another ten-by-five-fold interval and not a p-value for the concordance increment. A second null permuted training survival pairs on the five saved repeat-1 differential-expression fits, without refitting DESeq2, and cycled the same way across 200 simulations. The original 1,000 patient bootstraps and 20-repeat per-gene CV condition on the full-data shortlist and remain descriptive rather than selection-adjusted evidence.

The same patients, seeds, and event-stratified folds were then used to compare selectors. One training-only DESeq2 fit supplied six predictors for each held-out fold: the clinical model; the primary HYDRA gene; a survival-only gene held to the same high-confidence hazard gates without tumor-normal or GEO filters; a differential-expression-only gene held to the training DESeq2 thresholds without survival or GEO filters; a ridge Cox model on every training-fold reproducible DEG, with clinical covariates unpenalized and $\lambda_{\min}$ chosen by the training-only inner deviance cross-validation above [11]; and one expression-matched gene at or below the median residual correlation with the clinical martingale residuals. Survival-only false-discovery rates were computed as if every unscreened count-QC gene had $p = 1$, after fitting the top 2,000 residual correlations. The HYDRA gene was required to match the saved primary nested result. Intervals remain conditional on the saved predictions. This comparison does not replace the prespecified one-gene acceptance rule.

On those same folds we also added ClearCode34, the published 34-gene ccA/ccB score [3]. ccA genes were signed negative and ccB genes positive. Current symbols were taken from gene aliases where the published symbol has changed; one published symbol did not map. Expression was $\log_2$ counts per million, and the training fold supplied the mean and standard deviation. This arm does not select genes. Separately, on the previously inspected E-MTAB-1980 cohort, we fit in-sample Cox models for the study’s ccA/ccB label, for the highest-evidence primary gene present on that array (CRYL1; CYFIP2 did not map), and for a sign-weighted mean of the mapped high-confidence genes, each added to the limited clinical model. Those E-MTAB fits are not nested, and their in-sample calibration slopes are not held-out estimates.

2.3 Exploratory alternatives and biological checks

At the largest common list size of 22 genes, we compared the complete rule with survival-only, DE-only, each leave-one-GEO-out rule, and TCGA-expression-matched non-prognostic controls. GSE29609 and E-MTAB-1980 were previously inspected, so both evaluations are exploratory. We estimated the complete-rule

minus survival-only directional-rate difference and a 2,000-draw paired gene bootstrap in each cohort. The prespecified funnel criterion required an interval entirely above zero.

Candidate Cox models were refit on matched complete cases with published TCGA consensus tumor purity [1] or leave-one-candidate-out marker scores. Each score is the mean of sample-standardized marker expression, then standardized across samples. The sets are proximal tubule (AQP1, LRP2, CUBN, SLC5A2, ALDOB), endothelial (PECAM1, VWF, KDR, EMCN, TEK), immune (PTPRC, CD3D, CD8A, MS4A1, CD68), and stromal (COL1A1, COL1A2, DCN, LUM, ACTA2). A candidate that belongs to a set is omitted from that score. These are bulk marker scores, not measured cell proportions. Consensus purity is an inferred bulk estimate. Nonlinear expression and expression-by-log-time models were tested against the linear proportional-hazards fit. Human Protein Atlas normal-tissue single-cell profiles informed cell-source cautions but were not treated as ccRCC tumor single-cell validation.

TRACERx Renal provided primary-region expression and linked survival data from a pinned public repository [7]. One thousand repeats selected one region per patient. A fixed event-stratified 39-patient subset isolated region-choice variation; a separate scenario resampled which 39 patients entered each repeat. Check-Mate 025 supplementary RNA data supported post hoc continuous-expression-by-treatment Cox interactions for overall and progression-free survival, with FDR correction within endpoint and model [2].

3 Results

3.1 Corrected discovery and conditional sensitivity

533 patients to 517 nested

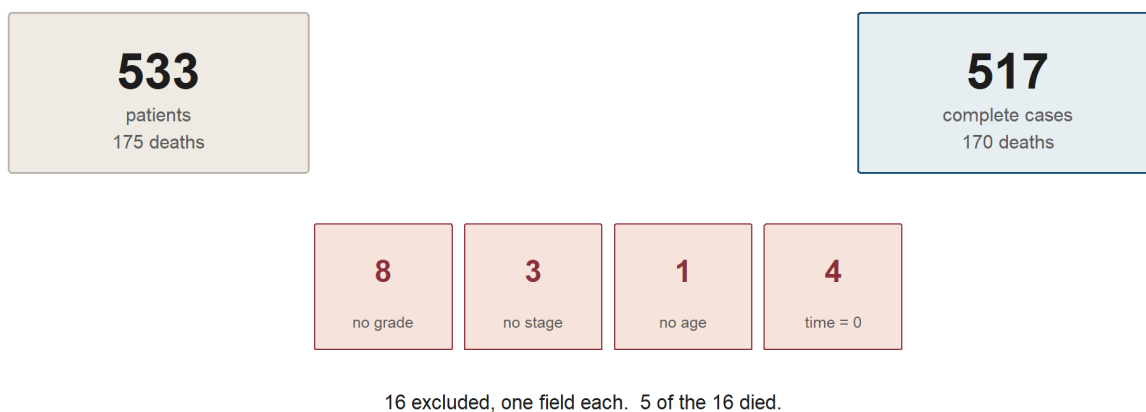


Figure 2: From 533 patients to the 517-patient nested cohort. The 16 exclusions are drawn separately: 8 lack grade, 3 lack stage, 1 lacks age, and 4 have survival time zero. 5 of those 16 died. Stage-only and grade-only sensitivities restore the grade-missing and stage-missing patients (525 and 520).

The original 541 tumor aliquots represented 533 patients; four patients contributed three aliquots each, accounting for the eight extra aliquots. No patient had more than one normal. Patient-level selection left 72 normals. Dropping the four multi-aliquot patients left 19 high-confidence genes, the first-barcode rule left 20, and summing tumor counts left 19. GRAMD1A, IFFO1, LTB4R, and RBM47 were absent from all three alternate sets. The deepest-library rule stays the primary rule. The unpaired GSE53757 gate kept 23 of the 23 primary genes and had Jaccard overlap 0.998 with the paired reproducible list. The paired DE fit included 72 patients and found 8,626 genes meeting the stated DE threshold. All 23 candidates retained their tumor-normal direction, while only 17 met the paired DE threshold; this is a sensitivity to the tumor-normal design, not an independent cohort. The primary analysis found 8,534 TCGA-significant mapped

genes, 3,323 reproducible DEGs, 1,186 main-model FDR associations, 1,117 sensitivity-pass genes, 538 strict candidates, and 23 high-confidence candidates. Relative to the previous 27-gene run, 3 genes entered and 7 left the shortlist. The gene-level columns for all 23 candidates are in Table 1. Every candidate has the same columns.

Table 1: Supplementary Table 1. Evidence matrix for the 23 high-confidence candidates. Every row uses the same columns. Bold marks the GSE29609 reversals DDC and TCIRG1. An asterisk marks a gene lost under every alternate aliquot rule. Direction is relative to the TCGA hazard. Nested folds count how often the gene was the selected gene. TRACERx is the fraction of fixed-patient region draws that kept the TCGA direction.

Gene	TCGA log ₂ FC	TCGA HR	GSE40435	GSE53757	GSE29609	E-MTAB-1980	Marker	Purity	Folds	TRACERx	Aliquot
CYFIP2	-2.28	0.64	yes	yes	opposite	absent	yes	yes	22	94%	yes
CRYL1	-1.07	0.63	yes	yes	opposite	same	yes	yes	9	100%	yes
HHLA2	3.12	0.63	yes	yes	opposite	same	yes	yes	11	100%	yes
DDC	-1.70	0.66	yes	yes	opposite	same	no	yes	1	96%	yes
KL	-1.64	0.66	yes	yes	absent	same	no	yes	1	37%	yes
RBM47*	-1.10	0.67	yes	yes	opposite	same	yes	yes	0	100%	no
PANK1	-1.49	0.62	yes	yes	opposite	same	yes	yes	0	84%	yes
GJB1	-1.47	0.65	yes	yes	same	same	no	yes	0	97%	yes
ACADM	-1.54	0.64	yes	yes	opposite	same	yes	yes	0	100%	yes
CLCN5	-1.43	0.66	yes	yes	opposite	same	no	yes	0	60%	yes
TMEM140	1.39	0.67	yes	yes	opposite	same	yes	yes	0	100%	yes
C1QTNF6	2.33	1.55	yes	yes	opposite	same	yes	yes	0	93%	yes
CADPS2	-1.63	0.66	yes	yes	same	same	yes	yes	0	100%	no
KNTC1	1.16	1.51	yes	yes	opposite	same	yes	yes	0	100%	yes
PODXL	-1.37	0.66	yes	yes	opposite	same	no	yes	0	100%	yes
HIBCH	-1.29	0.66	yes	yes	opposite	same	no	yes	0	99%	yes
TNIP1	1.05	0.66	yes	yes	opposite	same	yes	yes	0	69%	yes
TCIRG1	1.46	1.53	yes	yes	opposite	same	yes	yes	0	99%	yes
LTB4R*	2.59	1.51	yes	yes	same	same	yes	yes	0	39%	no
LRBA	-1.25	0.65	yes	yes	opposite	same	yes	yes	0	100%	yes
GRAMD1A*	1.32	1.50	yes	yes	same	same	yes	yes	0	98%	no
IFFO1*	1.78	1.50	yes	yes	absent	same	yes	yes	0	57%	no
TNFAIP2	1.29	1.53	yes	yes	same	opposite	yes	yes	0	64%	no

Figure 1 is that flow, ending in the held-out benchmark on the same 517 patients and splits.

The all-gene sensitivity modeled 32,192 count-QC genes and found 12,630 survival associations at global FDR below 0.05, illustrating how broad the outcome signal becomes without the upstream gate.

All 23 conditional bootstrap intervals excluded zero. The per-gene conditional CV showed positive mean concordance increments for 23 candidates, but both analyses start from genes chosen on the full TCGA outcome data. They cannot estimate the optimism of the complete selection process. The within-reproducible-gene overlap permutation did not show enrichment of the GEO-effect gate among survival-pass genes (empirical $p = 0.178$). It shuffles pass labels rather than refitting survival models, so it is a limited negative control.

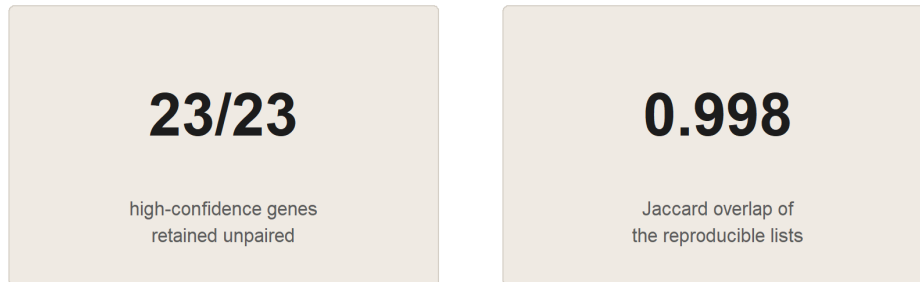
3.2 Selection-aware prediction and funnel comparison

The nested analysis selected no gene in 0 of 50 training folds, but selected 10 different genes overall; the most frequent appeared in 22 folds. Thus, universal selection did not imply a stable gene identity.

2 folds required a no-replacement retry after a DESeq2 outlier-replacement error. Mean out-of-fold concordance was 0.744 for the clinical model and 0.747 after adding the training-selected gene, an increment of 0.0039 (patient-bootstrap 95% interval -0.0088 to 0.0171). The mean gene-minus-clinical three-year Brier-score difference was -0.0009. Mean risk-score calibration slopes were 0.86 and 0.76, respectively; the latter was farther from one despite the slightly lower average Brier score. The predeclared prediction criterion failed. The clinical-only null selected a gene in 3 of 200 single-fold simulations. The permutation null, which keeps the training expression fits and scrambles training survival, selected a gene in 0 of 200 simulations. Neither frequency is a p-value for the ten-repeat mean increment.

On those splits, the benchmark HYDRA arm matched the primary selected gene in every fold and again changed concordance by 0.0039 (interval -0.0081 to 0.0183). Survival-only selection, which used 20 genes, changed concordance by -0.0011 (interval -0.0151 to 0.0139). Differential-expression-only selection concentrated on 5 genes and changed concordance by -0.0015 (interval -0.0120 to 0.0079). Both left the mean three-year Brier score worse than the clinical model. The expression-matched control, spread across

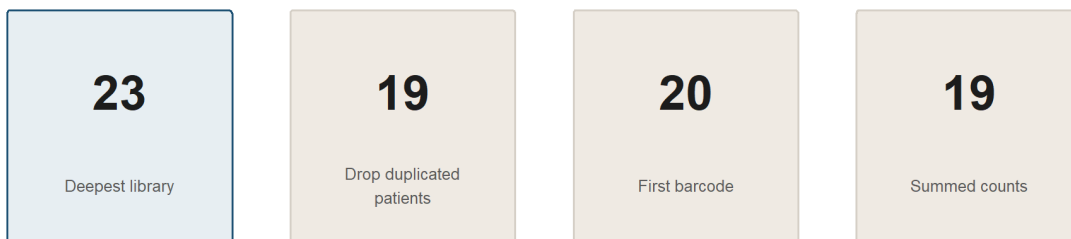
GSE53757



Paired versus unpaired. Adjacent rows do not prove patient identity.

Figure 3: Paired GSE53757 analysis versus the unpaired sensitivity. The unpaired gate retained 23 of 23 high-confidence genes. Jaccard overlap of the reproducible lists was 0.998. Adjacent-row pairing does not prove patient identity.

Aliquot rule



Lost under every alternate rule: GRAMD1A, IFFO1, LTB4R, RBM47

Figure 4: High-confidence genes under the primary deepest-library rule and three alternate aliquot rules. GRAMD1A, IFFO1, LTB4R, and RBM47 were lost under every alternate rule.

HYDRA-ccRCC Evidence Funnel

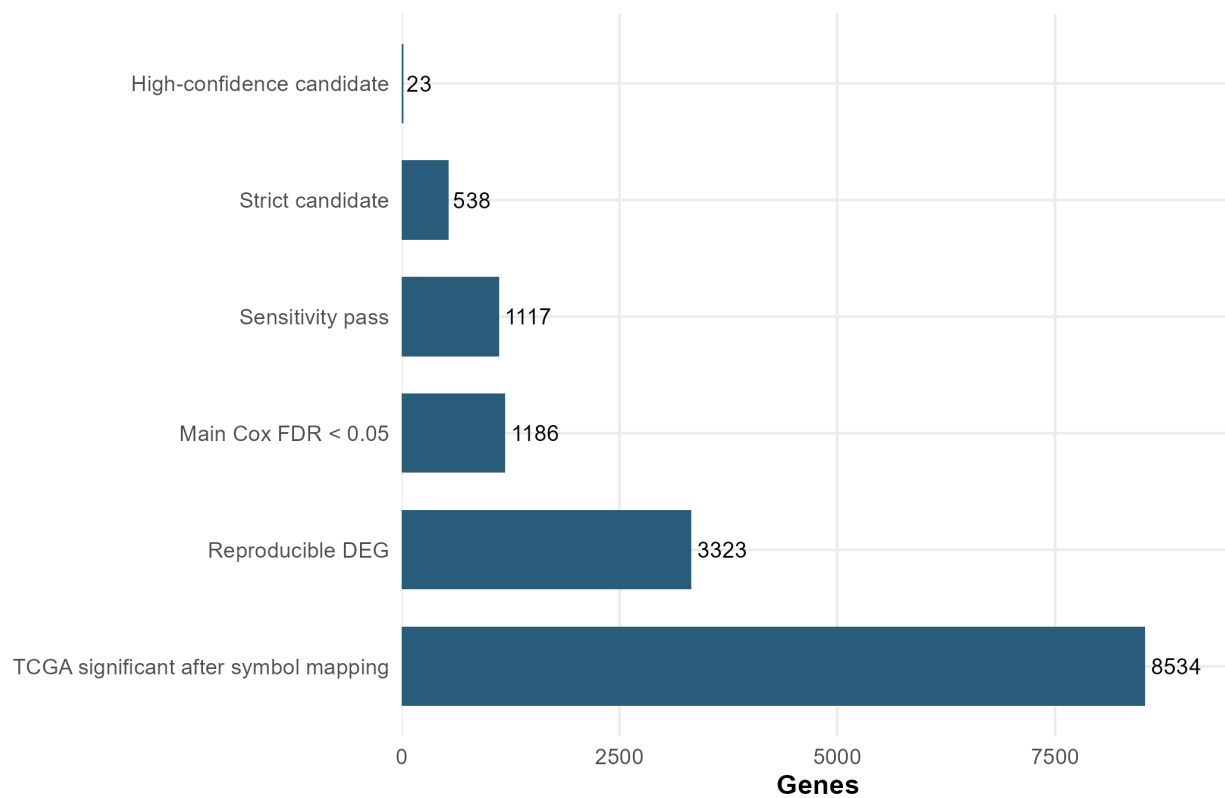


Figure 5: Corrected patient-level discovery funnel. The candidate set is defined entirely within TCGA and the two GEO expression cohorts.

Genes selected inside the nested folds

Fifty training folds. The acceptance test adds only the top gene.

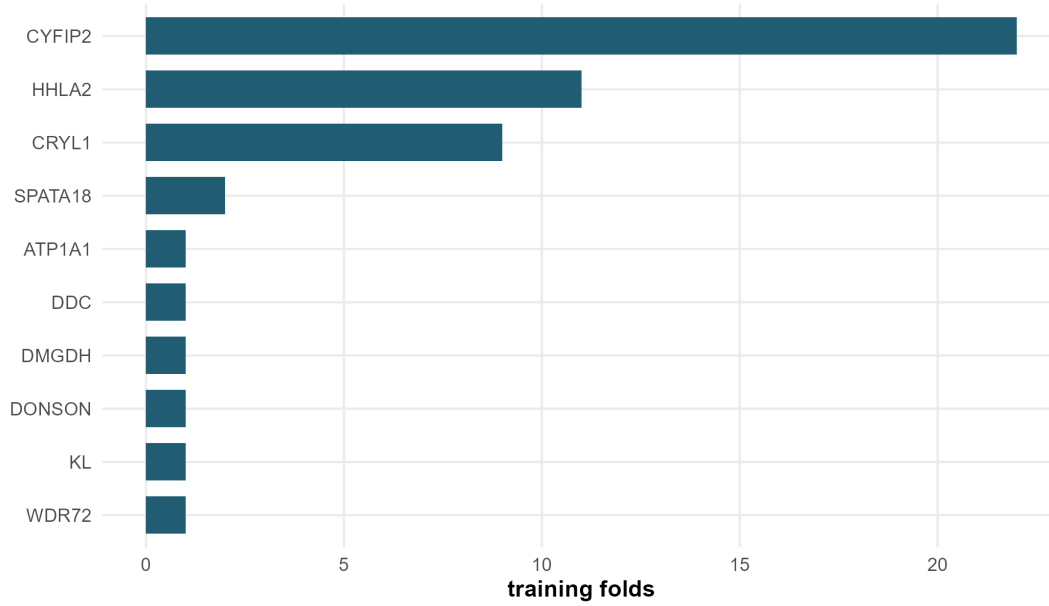


Figure 6: Training-fold selection frequency. Ten genes were chosen at least once. CYFIP2 was the top gene in 22 of 50 folds.

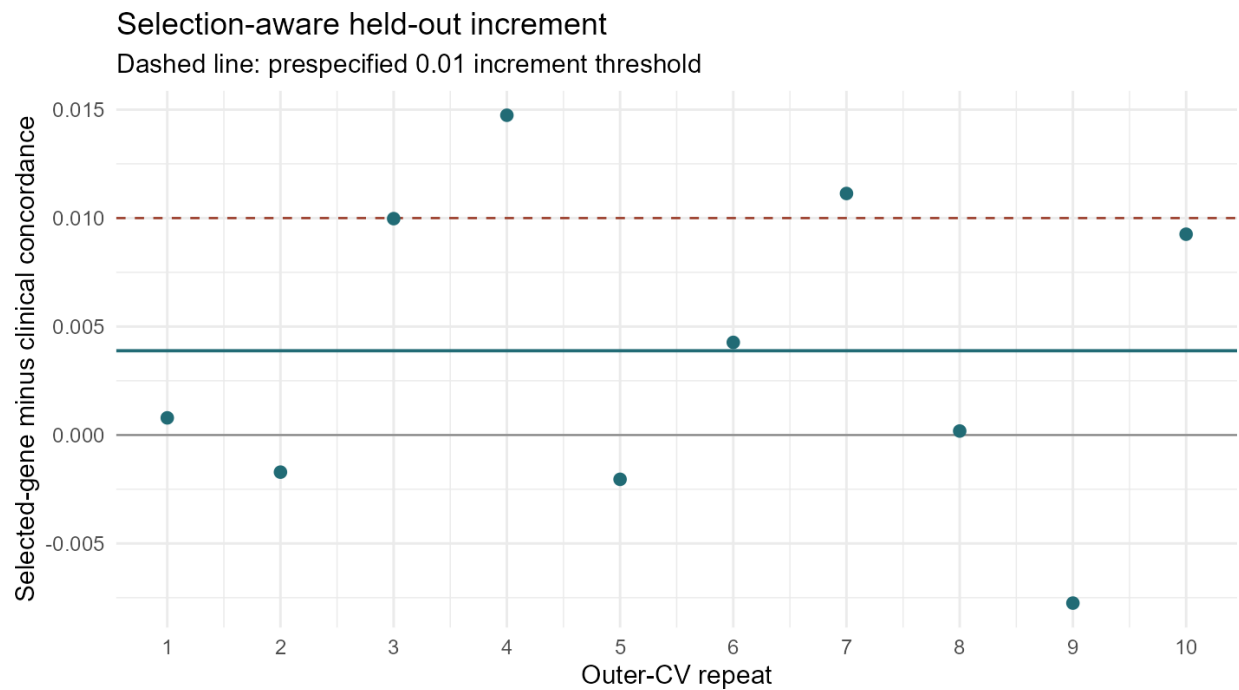


Figure 7: Concordance increment across ten complete outer-CV repeats. The dashed line marks the prespecified 0.01 threshold; the patient-bootstrap interval is reported in text.

44 genes, changed concordance by 0.0019 (interval 0.0001 to 0.0039). The interval sits above zero and remains well below 0.01, so a nonzero lower bound at this size is not a useful gain. Ridge regression kept the training-fold reproducible genes, on average 3,335 genes (range 3,067 to 3,447). Concordance rose by 0.0257 (interval 0.0116 to 0.0413), clearing the 0.01 line. The mean Brier difference was -0.0063 (interval -0.0141 to 0.0017), so the probability-score improvement is not stable under patient resampling. Calibration slopes were 0.86 for the clinical model, 0.76 after the single HYDRA gene, and 0.87 after the ridge model. The eligible set therefore carried held-out ranking information that none of the one-gene rules captured. The increment compares the joint ridge predictor with a separately fit clinical model. It does not partial out the change in the clinical coefficients, and the penalty does not identify which genes carry the gain. Size factors, the log expression, the nonzero training-SD filter, and λ were all taken from the outer training fold. The ridge arm is not a validated signature. Because the comparison was specified after the one-gene rule, it does not replace the failed one-gene criterion. It was not evaluated on GSE29609 or E-MTAB-1980, so the concordance gain is a within-TCGA ranking result. ClearCode34, using 33 mapped genes, changed concordance by 0.0129 (interval -0.0018 to 0.0287) and the three-year Brier score by -0.0046. The calibration slope was 0.83. ClearCode34 is a published comparator, not a rerun of the one-gene acceptance test. Its interval includes zero and values below 0.01, so it does not establish a robust gain of at least 0.01. Table 2 puts every arm in one place. Table 3 records the ridge specification.

Table 2: Held-out benchmark on the same 517 patients and splits. Changes are versus a separately fit clinical model. The one-gene acceptance test is the HYDRA row only. Ridge is a description of ranking information in the reproducible set. ClearCode34 is a published comparator. It was not scored against the one-gene acceptance rule.

Arm	ΔC	95% interval	Δ Brier	Slope
HYDRA	+0.0039	-0.0081 to +0.0183	-0.0009	0.76
Survival-only	-0.0011	-0.0151 to +0.0139	+0.0029	0.71
DE-only	-0.0015	-0.0120 to +0.0079	+0.0025	0.81
Matched control	+0.0019	+0.0001 to +0.0039	-0.0006	0.86
ClearCode34	+0.0129	-0.0018 to +0.0287	-0.0046	0.83
Ridge	+0.0257	+0.0116 to +0.0413	-0.0063	0.87

Table 3: Supplementary Table 2. Ridge specification. This fit is not the one-gene acceptance test. The penalty does not select a gene list.

Item	Definition
Predictor set	training-fold reproducible DEGs with nonzero training standard deviation; penalty factor 1; none are removed by the penalty
L2 penalty	0
Inner CV	five event-stratified folds of the training patients only
Lambda selection	lambda.min
Tuning metric	partial-likelihood deviance inside the training fold; not the C-index and not the one-standard-error rule
Training-only preprocessing	outer-training DESeq2 size factors, $\log_2(\text{count} / \text{size factor} + 1)$, and the nonzero training-SD gene filter are fixed before <code>cv.glmnet</code>
Unpenalized clinical covariates	age, sex, stage, and grade; penalty factor 0; factor levels from the complete-case cohort; values from the training rows only

On E-MTAB-1980, adding the study ccA/ccB label to the limited clinical model changed in-sample concordance by 0.0351 (interval -0.0483 to 0.1153). Adding CRYL1 changed it by 0.0034, and the signed panel of 22 mapped genes changed it by 0.0314. Those intervals include zero. The cohort has 23 deaths and was previously inspected.

The matched-list funnel criterion failed. In GSE29609 the complete-minus-survival-only directional-rate difference was 0.042 (paired gene-bootstrap 95% interval -0.194 to 0.270). The complete and survival-only

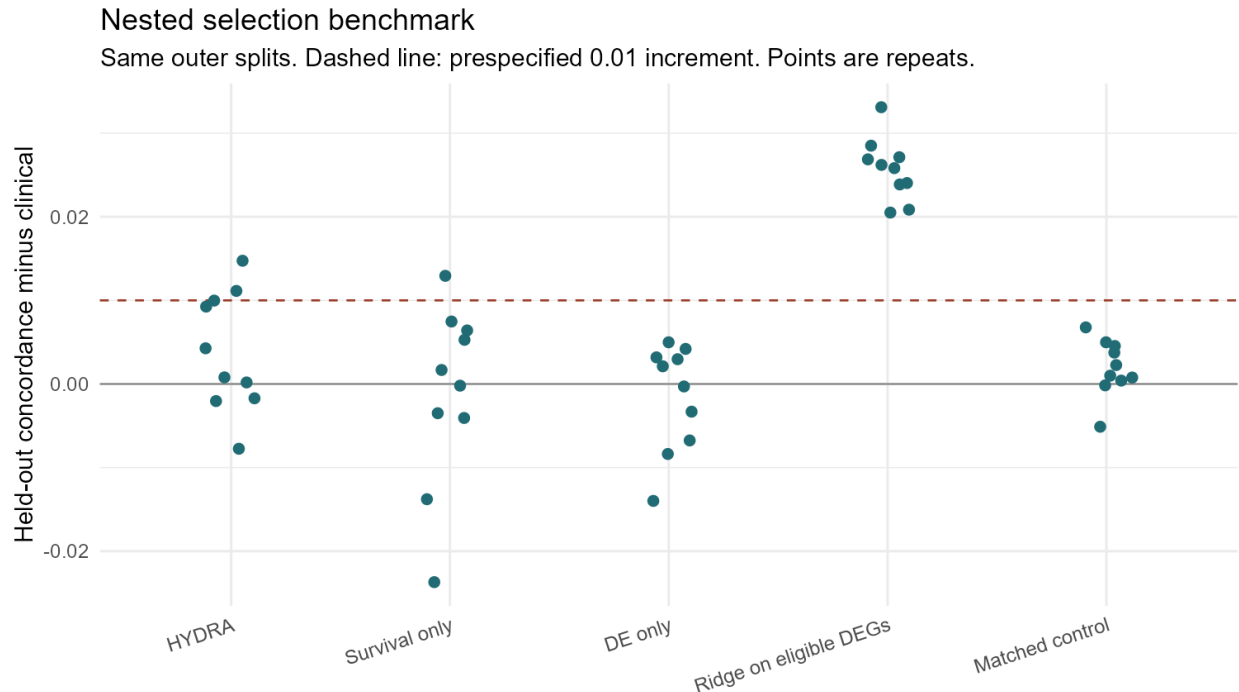


Figure 8: Repeat-level concordance increments for the five selectors fit inside the nested benchmark. ClearCode34 is on the master figure. The dashed line is the prespecified 0.01 threshold.

lists mapped 20 and 19 genes there, respectively, so the observed difference compares proportions with distinct denominators. In E-MTAB-1980 both lists had 100% direction agreement among mapped genes (21 and 18, respectively), leaving no directional separation. Matching at 22 selected genes was necessary because one leave-one-GEO-out rule yielded only 22 eligible genes. The comparison reuses already inspected external outcomes and is not confirmatory.

3.3 External disagreement and composition

ACADM, CRYL1, and DDC are three evidence profiles, not a ranked set of winners. ACADM is a concordant metabolic association: downregulated in TCGA and both GEO cohorts, with a protective TCGA hazard ratio. CRYL1 has a stronger main-model FDR than ACADM and is the high-confidence gene scored on E-MTAB-1980, because the nested procedure’s most frequent gene, CYFIP2, did not map to that array. DDC has a large GEO downregulation and a protective TCGA association, then reverses in GSE29609 at FDR below 0.05. The matrix records the same columns for the other genes. These are candidate associations, not mechanisms or clinical biomarkers.

GSE29609 contained 39 patients and 17 deaths. It mapped 21 primary candidates, of which 5 retained the TCGA hazard direction; 4 had nominal opposite-direction associations. DDC and TCIRG1 were reversed at FDR below 0.05. E-MTAB-1980 contained 101 patients and 23 deaths, mapped 22 candidates, retained the TCGA direction for 21, and gave 12 strict same-direction unadjusted and limited-adjustment FDR results. These cohorts do not support a claim of uniform external replication. GSE29609 has 17 deaths.

Matched-patient marker-score adjustment removed FDR support for 6 candidates: CLCN5, DDC, GJB1, HIBCH, KL, PODXL. Direct consensus-purity adjustment retained direction and FDR support for all 23. The two adjustments do not agree, so composition control is model-dependent. Consensus purity remains an inferred bulk estimate rather than a tumor-cell measurement.

After correction for 23 tests, 0 nonlinear-expression comparisons and 0 expression-by-time comparisons had FDR below 0.05. Normal-tissue HPA profiles do not identify ccRCC tumor-cell origin.

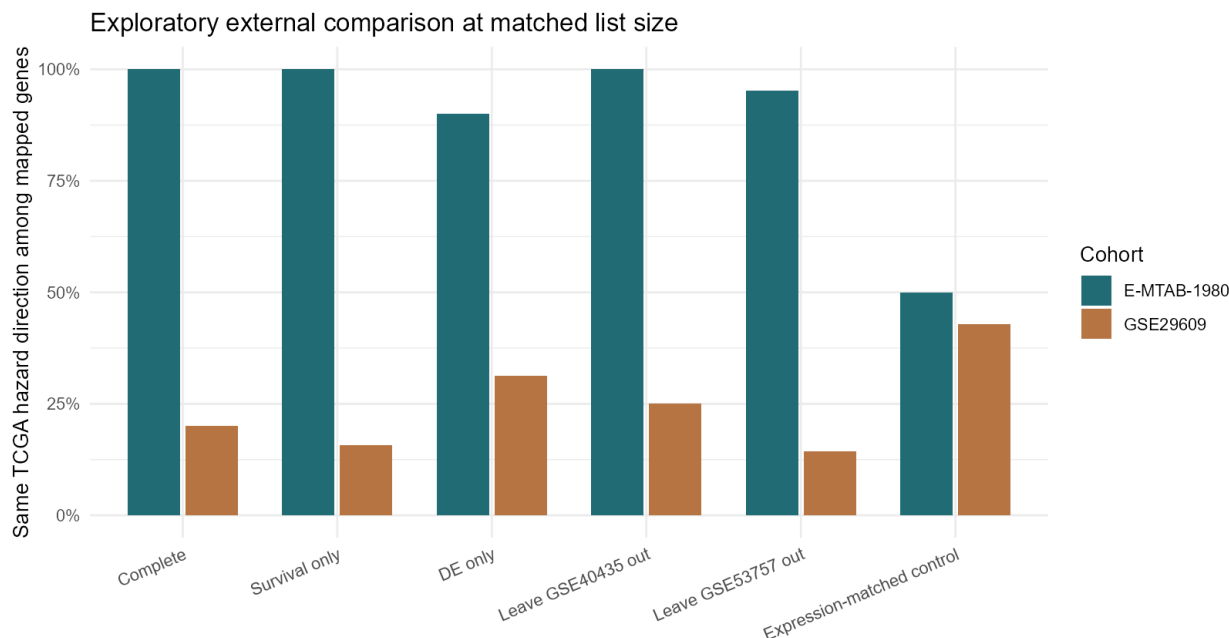


Figure 9: Same-direction agreement among mapped genes for six equal-sized lists in two previously inspected cohorts. Different platform coverage and event counts limit comparison. This comparison is not confirmatory.

3.4 Spatial sampling and randomized treatment

TRACERx linked 186 primary regions from 73 patients, including 16 deaths. A fixed event-stratified subset of 39 patients matches the size of GSE29609 and isolates region choice. Across genes, agreement was 98.5% [76.6–100.0] when those patients were fixed and only the chosen region varied, versus 86.3% [77.8–93.1] when both patient membership and region varied. Each smaller subset had 9 deaths. That event count makes this a sampling-variation check. It cannot attribute GSE29609’s reversals to spatial sampling alone.

The CheckMate 025 RNA subset included 250 patients (120 nivolumab and 130 everolimus); all 23 candidates were assessed. No expression-by-treatment interaction survived FDR correction for overall or progression-free survival in either the randomized or limited-adjustment model. Nominal interactions remain exploratory and do not establish treatment selection.

4 Discussion and limitations

The one-gene acceptance test failed. The point estimate is 0.0039. The patient-bootstrap interval, conditional on the saved predictions, runs from -0.0088 to 0.0171: it crosses zero, and its upper end is above 0.01, but the estimate itself is not. That failure does not make the 23 associations null. Labels stay in this order: a candidate prognostic association, then external agreement, then held-out discrimination, then a treatment interaction. Prognostic means an association with outcome. A treatment interaction is a separate question, and CheckMate 025 did not support one.

Ridge is not that acceptance test. On the same splits it shows held-out ranking information in the reproducible gene set (0.0257, interval 0.0116 to 0.0413). The Brier interval still crossed zero, the gain was not taken outside TCGA, and the penalty does not name a signature. ClearCode34 is a published comparator. Its interval includes zero and values below 0.01. It was not scored as the one-gene rule.

The external cohorts do not support a claim of uniform external replication. GSE29609, with 17 deaths, kept the TCGA direction for 5 of 21, and DDC and TCIRG1 reversed. E-MTAB-1980 kept it for 21 of 22, with strict support for 12. Both cohorts had already been inspected. TRACERx agreement across genes was 98.5% [76.6–100.0] with patients fixed and 86.3% [77.8–93.1] when membership was redrawn. Those subsets have 9 deaths. Marker scores and consensus purity disagree. The aliquot edge is 19, 20, and 19 under the

External survival does not copy TCGA

Shaded rows are DDC and TCIRG1. GSE29609 has 17 deaths; E-MTAB-1980 has 23.

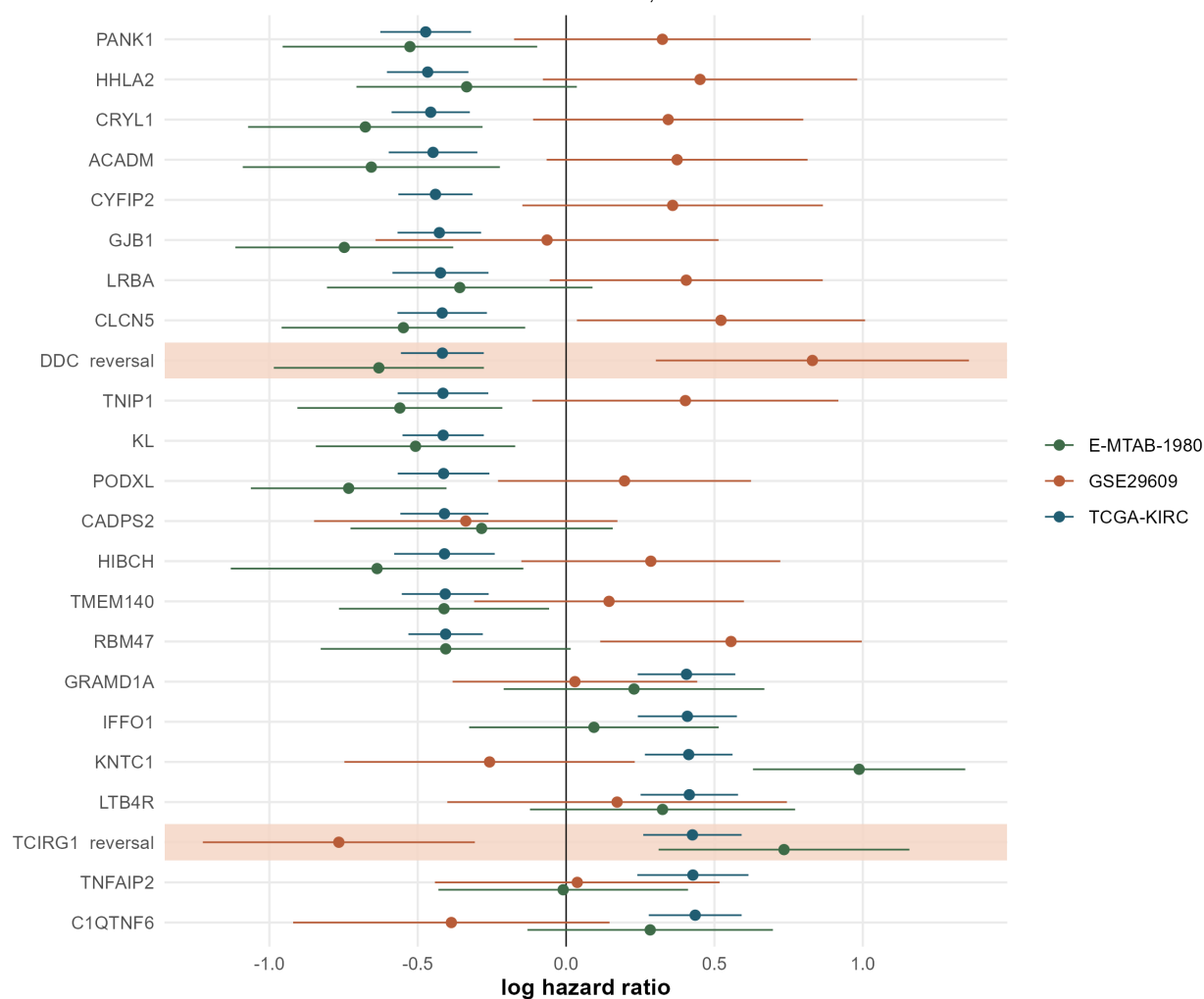


Figure 10: Log hazard ratios for the high-confidence genes in TCGA-KIRC, GSE29609 (17 deaths), and E-MTAB-1980 (23 deaths). Shaded rows are the DDC and TCIRG1 reversals. A shared direction is not an interval that excludes zero.

External direction

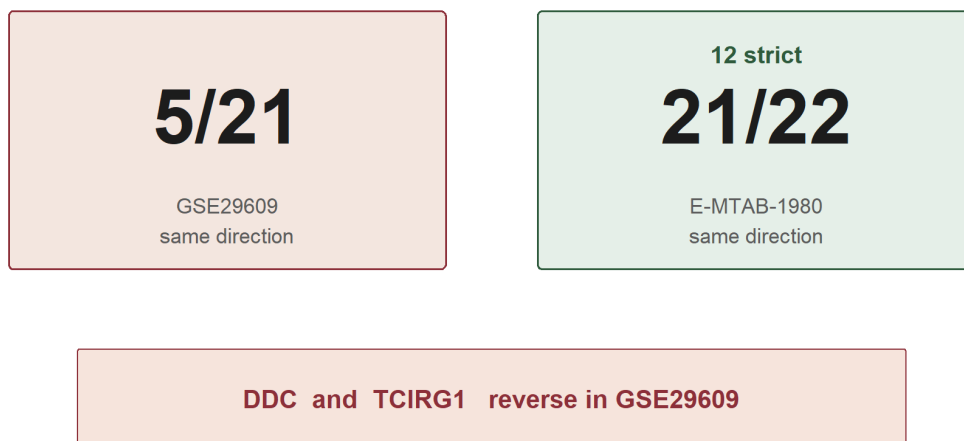


Figure 11: Same-direction counts among mapped high-confidence genes. GSE29609 retained the TCGA direction for 5 of 21. E-MTAB-1980 retained it for 21 of 22, with strict support for 12. DDC and TCIRG1 reverse in GSE29609.

three alternate rules, against 23 for the deepest library. GRAMD1A, IFFO1, LTB4R, and RBM47 leave under every alternate rule. The unpaired GSE53757 gate kept 23 of 23 genes (Jaccard 0.998) and does not prove that adjacent rows are patient pairs. None of this is a clinical action.

This is a retrospective public-data analysis, and no untouched external cohort is available. The matched-list interval resamples genes whose results can be dependent, so it is descriptive. Consensus purity and marker scores are imperfect composition measures. Human Protein Atlas profiles describe normal tissue. RNA associations do not establish a mechanism or a therapeutic target.

Data and reproducibility

Public source identifiers, the pinned TRACERx commit, cached-input hashes, and package versions are recorded in the repository. The original retrieval dates for cached inputs are unknown. Core result counts and estimates are generated from result tables by `analysis/25_paper_numbers.R` into `paper/results_macros.tex`; the pipeline and PDF build are separate reproducible commands. This manuscript is an internal AI-assisted development draft for independent verification before submission.

Ethics, funding, and AI assistance

This secondary analysis used public de-identified data and involved no new recruitment or intervention. No external funding was reported. OpenAI Codex assisted with the audit, code, and development manuscript; the student researcher must independently verify every statistical, scientific, and citation claim before any submission.

Acknowledgments

The previous development draft thanked Levi Waldron, Michael Love, Philip Saylor, Samra Turajlić, and David McDermott for feedback that shaped the analysis. The author remains responsible for the implementation and interpretation; these acknowledgments require author verification before submission.

Composition adjustment of the 23 candidates

Triangles lose FDR support after marker-score adjustment. The line is 0.05.

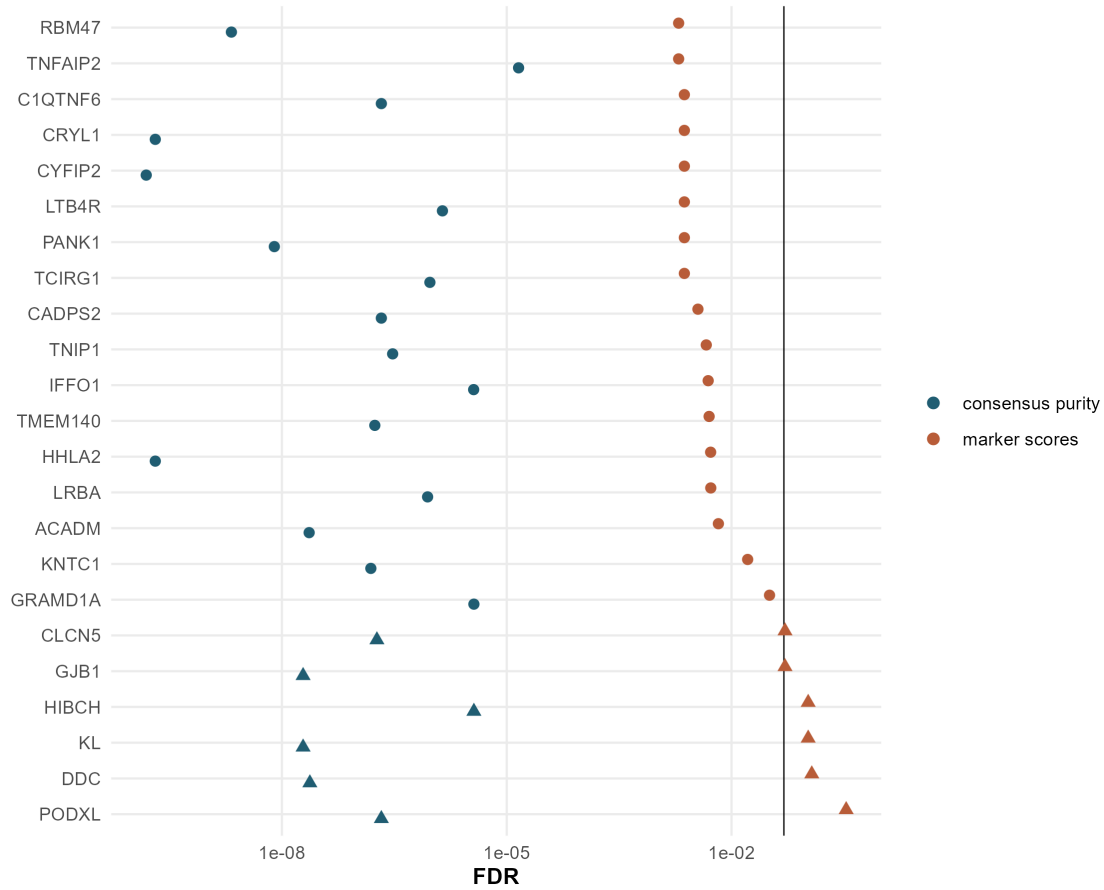


Figure 12: Marker-score adjustment and consensus-purity adjustment, shown together. Triangles mark the six genes that lose FDR support under the marker scores: CLCN5, DDC, GJB1, HIBCH, KL, PODXL. Consensus purity retains all 23. The vertical line is 0.05.

TRACERx direction

Panel labels are the median and interquartile range across genes.

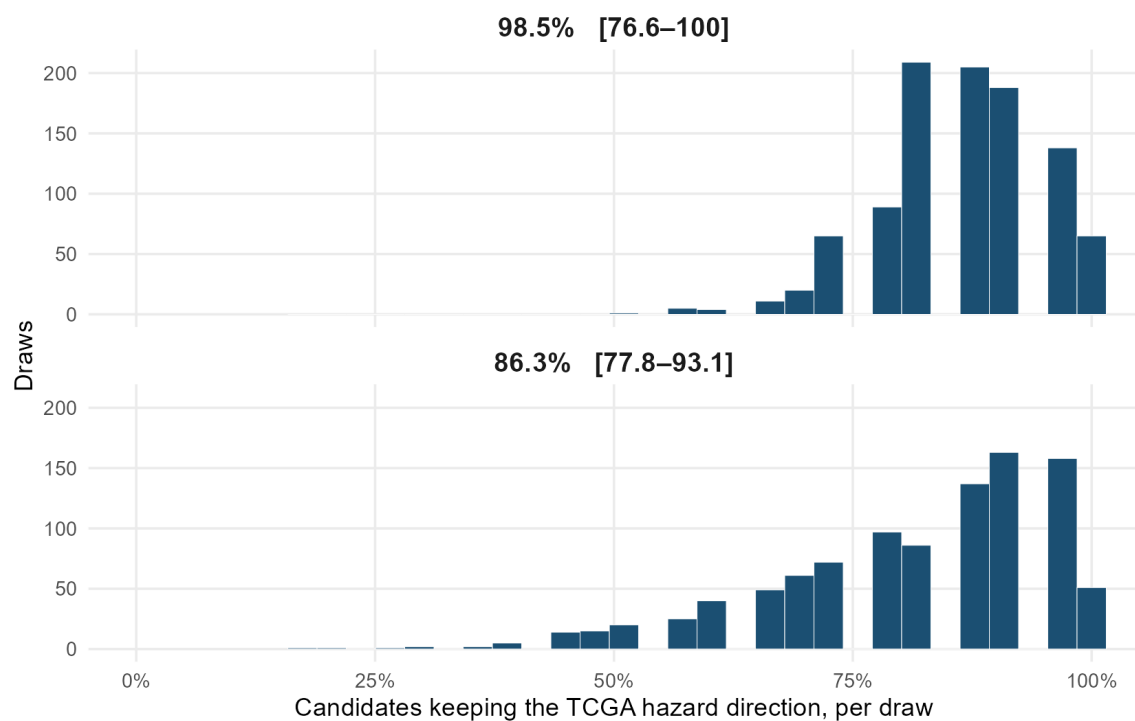


Figure 13: Fraction of high-confidence candidates that keep the TCGA hazard direction in 1,000 TRACERx draws. Across genes, agreement was 98.5% [76.6–100.0] with the 39 patients fixed, and 86.3% [77.8–93.1] when patient membership was also redrawn. Both subsets have 9 deaths. The brackets are interquartile ranges across genes.

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